MIL-C-52705A
26 May 1977
SUPERSEDING
MIL-C-52705
9 April 1971

MILITARY SPECIFICATION

COUPLINGS: HOSE, STEAM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers steam hose couplings.

1.2 Classification. The couplings shall be of the following types and sizes, as specified (see 6.2):

Type A - Expansion ring coupling with slotted swivel.
Size - 3/4 inch.

Type B - Ribbed shank coupling with hexagonal swivel.
Sizes - 1/2 inch, 3/4 inch, 1 inch, 1-1/4 inch, and 1-1/2 inch.

Type C - Ribbed shank coupling with wing nut swivel.
Sizes - 1/2 inch, 3/4 inch, 1 inch, 1-1/4 inch, 1-1/2 inch, 2 inch, and 2-1/2 inch.

1.2.1 Part number. The military specification part number for the couplings described in this specification will be identified as shown in 6.3.

FSC 4730

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-B-626	- Brass, Leaded and Nonleaded: Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar and Strip).
QQ-C-390	- Copper Alloy Castings (Including Cast Bar).
QQ-I-666	- Iron, Malleable, Ferritic, for Castings.
QQ-P-416	- Plating, Cadmium (Electrodeposited).

MILITARY

MIL-P-775	- Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles, and Strainers.

STANDARDS

FEDERAL

FED. TEST METHOD STD. No. 151	- Metals; Test Methods.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The couplings shall be as shown in Figure 1 through Figure 4 and as specified herein.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.2.1 Castings. Cast parts of couplings shall conform to QQ-C-390, Alloy No. 852, 854, or 857, or QQ-B-626, Alloy No. 260 or Composition 11, except that Alloy No. 268 may be used for swivels. Type C coupling components shall be malleable iron conforming to QQ-I-666, Grade II. Castings with holes are not acceptable.

3.2.2 Expansion rings. Expansion rings shall be made from seamless drawn tubing conforming to either of the compositions prescribed in Table I.

TABLE I. Composition.

Copper Percent	Tin Percent Maximum	Lead Percent Maximum	Iron Percent Maximum	Zinc Percent
79-86	-	0.2	0.05	Remainder
59-68	1.5	3.5	0.20	Remainder

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3.2.3 Rubber washers. The rubber washers shall be molded from heat-resistant and water-resistant synthetic rubber stock and shall conform to the dimensions shown in Figures 2 and 3.

3.2.4 Type B coupling parts and clamps. All Type B coupling parts and clamps shall be brass conforming to QQ-C-390, Alloy No. 852.

3.3 Finish. Outside surfaces of sandcastings shall be made smooth by shotblasting or sandblasting. The inside surfaces of all parts, except the corrugations in male and female sections, shall be machine finished.

3.3.1 Finish on Type C couplings. Unless otherwise specified (see 6.2), Type C coupling components, except washers and soft metal contact surface inserts, shall be cadmium-plated in accordance with the requirements of QQ-P-416, Class 3, Type I.

3.4 Corrugations. The corrugations in male and female sections shall be left rough so as to bind the hose.

3.5 Threads. Type A and Type B couplings shall be threaded NPSH and Type C couplings shall be threaded NPT. All threads shall be in accordance with NBS Handbook H28.

3.6 Parts. A complete set of couplings of the sizes specified (see Figures 2, 3, and 4) shall consist of the following parts:

Type A	- Part 7	-	Female section.
	- Part 8	-	Male section.
	- Part 3	-	Swivel of female section.
	- Part 4	-	Expansion rings (one each for male and female sections).
	- Part 5	-	Rubber washer fitting recess in swivel.
Type B	- Part 1	-	Female section.
	- Part 2	-	Male section.
	- Part 3	-	Swivel of female section.
	- Part 4	-	Clamps for attaching hose.
	- Part 5	-	Washer fitting recess in swivel.
Type C	- Part 9	-	Coupling.

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3.7 Hydrostatic pressure. The couplings shall be capable of withstanding a hydrostatic pressure of 300 pounds per square inch (psi) when tested in accordance with 4.4.2.1.

3.8 Identification marking. The couplings shall be identified in accordance with MIL-STD-130.

3.9 Workmanship. Material shall be uniform in quality, clean, smooth, sound, and free from foreign material, blowholes, slivers, laps, cracks, seams, scale, burrs, buckles, damaged ends and edges, and other defects which will detrimentally affect the serviceability of the couplings.

3.9.1 Castings. Castings shall be sound and free from patching, misplaced coring, warping, and other defects which might render the casting unsound for its intended use.

3.9.2 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable figures.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

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4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.2 Examination.

4.3.2.1 Individual. Each coupling shall be examined for the critical defect specified in 4.4.1. Presence of a critical defect shall be cause for rejection.

4.3.2.2 Samples. Samples selected in accordance with 4.3.1 shall be examined for the major and minor defects specified in 4.4.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2.1 and 4.4.2.2. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The coupling assembly shall be examined as specified herein for the following defects:

Critical

1. Holes in casting.

Major

101. Dimensions not as specified.
102. Material not as specified.
103. Incorrect or missing parts.
104. Finish not as specified.
105. Corrugations not as specified.
106. Threads not as specified.
107. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or illegible.

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4.4.2 Test.

4.4.2.1 Hydrostatic test. Subject the complete coupling assembly to a hydrostatic pressure of 300 psi for not less than 30 seconds. Examine the coupling assembly for leakage, rupture, and yielding. Leakage, rupture, or yielding shall constitute failure of this test.

4.4.2.2 Spectro-chemical analysis. Not less than a 2-ounce metal specimen shall be taken from each lot of couplings for analysis in accordance with FED. TEST METHOD STD. No. 151, Method 112. Failure to meet the requirements of 3.2.2 shall constitute failure of this test.

4.5 Inspection of packaging. The preservation, packing, and military marking shall be examined to determine compliance with the referenced documents in Section 5 of this specification. The commercial marking shall be examined to determine compliance with the requirements specified in Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level C, or Commercial, as specified (see 6.2).

5.1.1 Level A. The couplings shall be preserved in accordance with the Level A preservation and packaging requirements of MIL-P-775.

5.1.2 Level C/Commercial. The couplings shall be preserved in accordance with the Level C/Commercial preservation and packaging requirements of MIL-P-775.

5.2 Packing. Packing shall be Level A, Level B, Level C, or Commercial, as specified (see 6.2).

5.2.1 Level A. The couplings, preserved as specified in 5.1, shall be packed in accordance with the Level A packing requirements of MIL-P-775.

5.2.2 Level B. The couplings, preserved as specified in 5.1, shall be packed in accordance with the Level B packing requirements of MIL-P-775.

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5.2.3 Level C/Commercial. The couplings, preserved as specified in 5.1, shall be packed in accordance with the Level C packing requirements of MIL-P-775.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with the requirements specified in MIL-P-775.

5.3.2 Commercial. Marking for commercial packaging shall reflect as a minimum, the item name, national stock number, quantity, contract or purchase order number, and appropriate address markings on the shipping container. The marking shall be legible and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The couplings are intended for use in joining hose in steam systems.

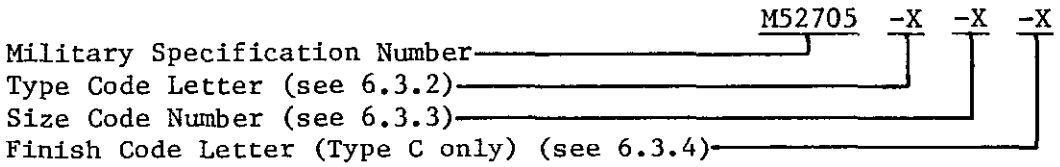
6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size coupling required (see 1.2 and 3.6).
(c) When Type C coupling components shall not be cadmium plated (see 3.3.1).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Definitive military specification part number. The military specification part number is a definitive part number which corresponds to the type designation and the size of the couplings covered by this specification and defines the requirements of the options presented under this specification. The specification number, type coupling code letter, size code number, and finish code letter are combined to form the difinitive military specification part number.

6.3.1 Cataloging data. For cataloging purposes, part numbers for the couplings are assigned as follows:

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6.3.2 Type coupling designation. The coupling type is identified by a single alphabetic character (see Table II).

TABLE II. Type code letter.

Code Letter	Type
A	A
B	B
C	C

Type A - Expansion ring coupling with slotted swivel.
Type B - Ribbed shank coupling with hexagonal swivel.
Type C - Ribbed shank coupling with wing nut swivel.

6.3.3 Size. The size of the coupling is identified by a single digit number (see Table III).

TABLE III. Nominal size code number.

Code Number	Nominal Size (in inches)
1	1/2
2	3/4
3	1
4	1-1/4
5	1-1/2
6	2
7	2-1/2

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6.3.4 Finish. Finish for the Type C coupling shall be identified by a single alphabetic letter (see Table IV).

TABLE IV. Finish code letter.

Code Letter	Finish
A	Without plating
B	With cadmium plating

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activities:
Army - GL
DLA - CS

Preparing activity:
Army - ME

Project 4730-0641

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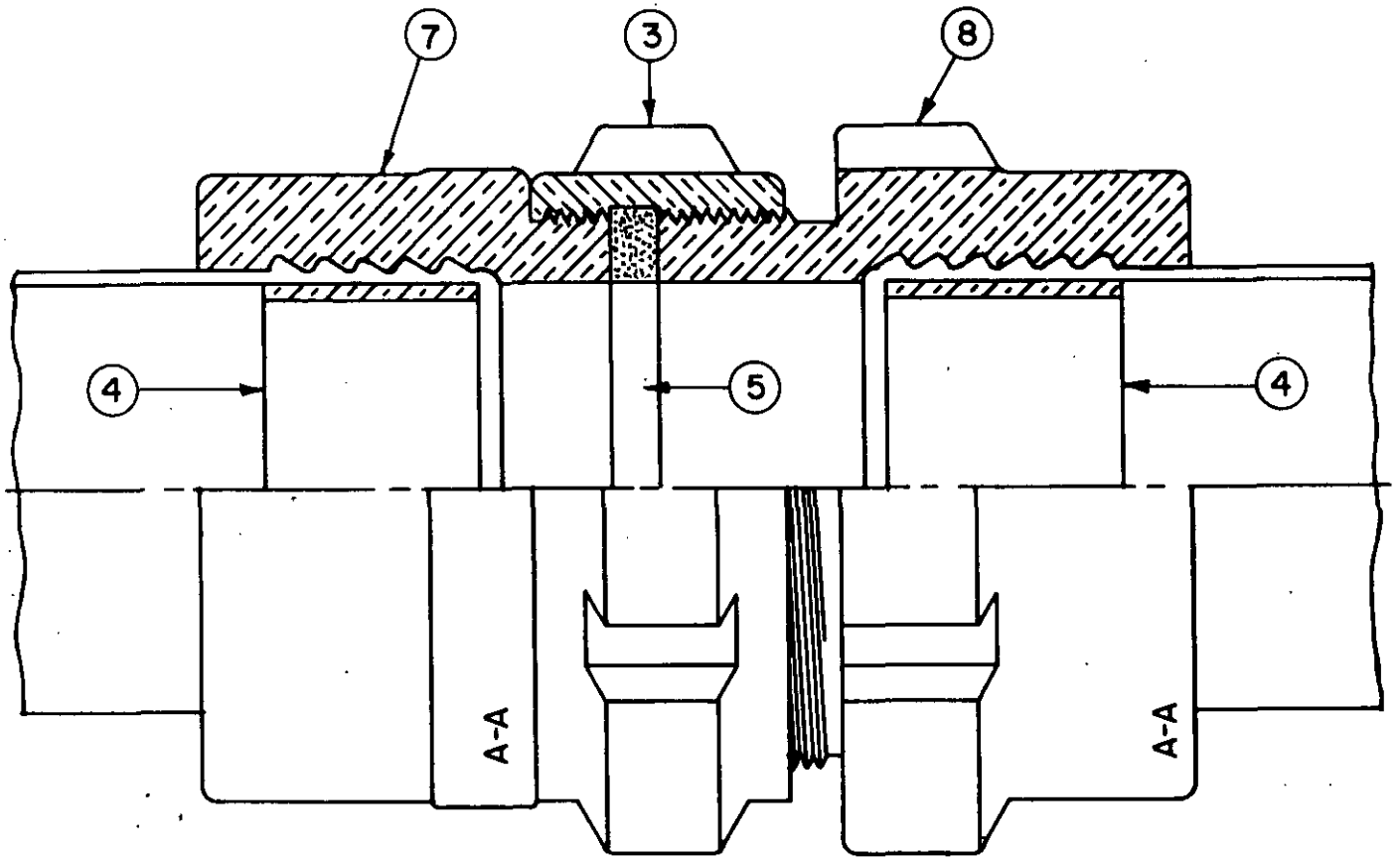
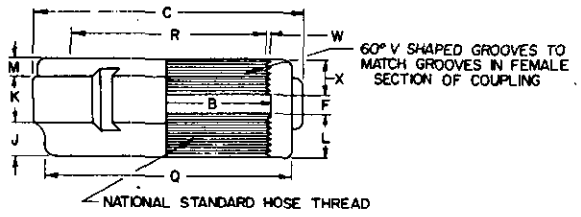
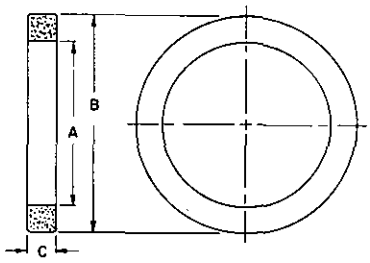


Figure 1. COUPLING ASSEMBLY-TYPE A.

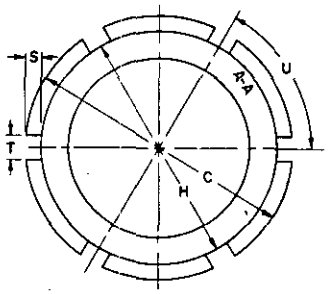
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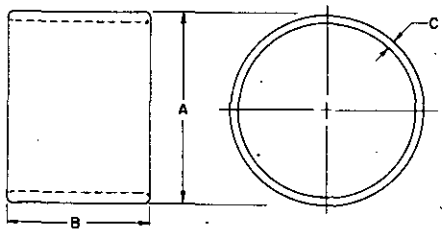




RUBBER WASHERS
PART 5

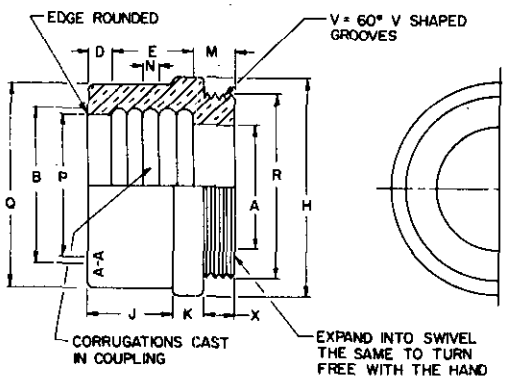


SWIVEL PART 3



EXPANDING RING
PART 4
EDGES WELL ROUNDED AND FREE FROM BURRS

COUPLING FEMALE SECTION
PART 7



COUPLING MALE SECTION
PART 8

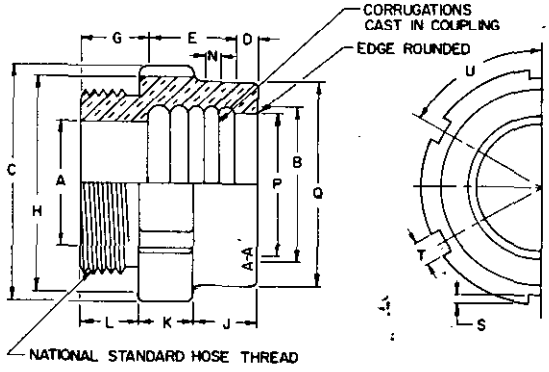


TABLE OF PARTS FOR (3/4 INCH) STEAM HOSE COUPLING

Part No.	A	B	C	D	E	F	G	H	J	K	L
7 and 8	7/8	-------	2-1/4	3/16	1-1/2	0	15/16	2	1-1/4	5/8	3/4
3	------	1-7/16	2-1/4	------	------	1/4	------	2	1/4	5/8	1/2
4	3/4	1-1/4	0.04 (min.). 0.05 (max.).	------	------	------	------	------	------	------	------
5	13/16	1-13/32	3/16 1/8 (min)	------	------	------	------	------	------	------	------

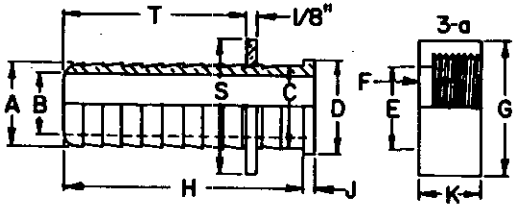
Part No.	M	N	P	Q	R	S	T	U	V	W	X
7 and 8	1/2	3/16	1-1/2	1-15/16	1-5/16	1/8	1/4	60°	4	1/16	5/16
3	3/16	-------	------	1-15/16	1-5/16	1/8	1/4	60	4	1/16	5/16

FIGURE 2. Steam hose coupling Type A.

X-2002A

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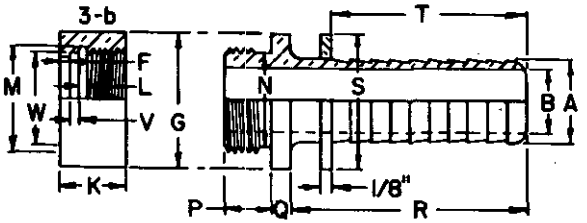
PART No. 1



PART No. 3



PART No. 2



PART 3-a USED ON SIZES 7/16" TO 1" INCLUSIVE
PART 3-b USED ON SIZES 1-1/4" & 1-1/2"

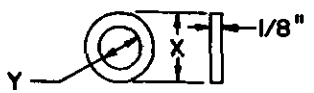
TOLERANCES
ROUGH CASTINGS INSIDE - 1/64" OUTSIDE + 1/64
FINISH ± .005"

SIZE	A	B	C	D	E	F	G	H	J	K	L	M
1/2	9/16	3/8	19/32	23/32	39/64	3/32	1-3/32	2-3/4	3/32	21/32		
3/4	13/16	9/16	53/64	15/16	27/32	1/8	1-5/16	2-15/16	3/32	11/16		
1	1-3/64	13/16	1-1/16	1-11/64	1-5/64	1/8	1-21/32	3	3/32	13/16		
1-1/4	1-5/16	1-1/16	1-11/32	1-31/64	1-23/64	1/8	2	3-7/16	1/8	7/8	3/16	1-11/16
1-1/2	1-9/16	1-5/16	1-19/32	1-47/64	1-39/64	1/8	2-7/32	3-11/16	1/8	7/8	3/16	1-15/16

SIZE	N	P	Q	R	S	T	U	V	W	X	Y	
1/2	RELIEVE AT LEAST TO BOTTOM OF THREAD.	7/16	3/16	2-3/8	7/8	2	1/2			47/64	1/2	
3/4		1/2	3/16	2-13/16	1-5/16	2-1/4	3/4			15/16	11/16	
1		9/16	1/4	3	1-19/32	2-1/4	1			1-3/16	15/16	
1-1/4		5/8	1/4	3-1/8	1-7/8	2-1/2	1-1/4	1/8	1-1/2	1-5/8	1-1/4	
1-1/2		5/8	1/4	3-9/16	2-11/32	2-3/4	1-1/2	1/8	1-3/4	1-7/8	1-1/2	

* U = SIZE OF HOSE THREADS

PART No. 5



PART No. 4

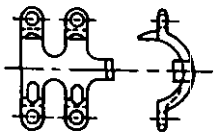
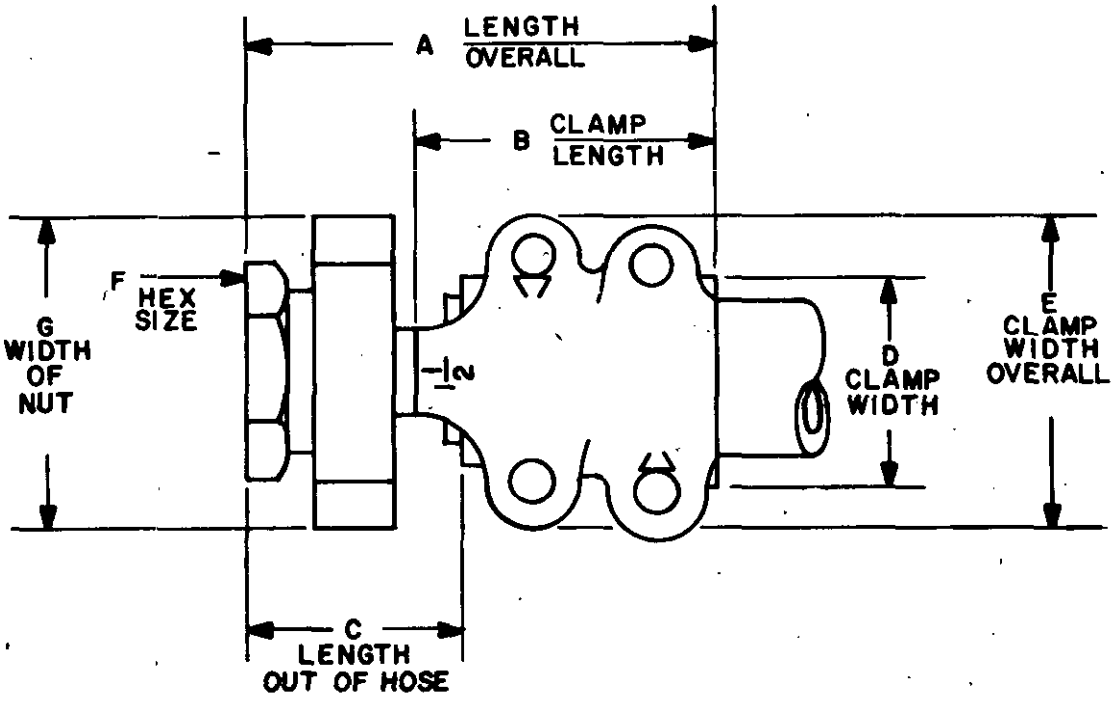


FIGURE 3. Steam hose couplings Type B.

X-2003A

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PART 9

NPT	HOSE SIZE	A	B	C	D	E	F	G
1/2	1/2	3-11/16	1-13/16	2-1/16	1-5/16	2-1/8	1-5/16	2-5/16
3/4	3/4	4-7/8	2-3/4	2-1/2	1-13/16	2-3/4	1-7/8	3-9/16
1	1	5-1/4	3-1/8	2-1/2	2-5/16	3-3/8	1-7/8	3-9/16
1-1/4	1-1/4	6-1/2	3-3/4	3-3/8	4	5	2-7/16	4-1/4
1-1/2	1-1/2	6-5/8	3-7/8	3-3/8	4-1/8	5-1/8	2-9/16	4-1/4
2	2	7-1/4	4-1/4	3-3/4	5-1/16	6-1/16	3-1/16	5-5/8
2-1/2	2-1/2	8-1/2	5	4-1/4	6-3/8	7-1/4	4	6-3/4

NPT	Shank dia in Hose	Shank lg in Hose	Wt in lbs of Coupling
1/2	17/32	1-5/8	1.0
3/4	25/32	2-3/8	2.8
1	1-1/32	2-5/8	3.6
1-1/4	1-9/32	3-1/4	6.7
1-1/2	1-9/16	3-3/8	6.9
2	2-1/32	3-15/16	11.0
2-1/2	2-9/16	4-1/2	18.7

FIGURE 4. Steam hose coupling-Type C. X-2082A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-C-52705A Couplings: Hose, Steam

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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